

4.2	Geology and hydrostratigraphy	507
4.3	Dewatering and depressurisation	509
4.4	Conceptual hydrogeological model	511
4.5	Discussion	512
5	Jwaneng Diamond Mine South-east wall	513
5.1	Background	513
5.2	Geology and hydrostratigraphy	513
5.3	Dewatering program	513
5.4	Slope depressurisation	516
5.5	Discussion	519
6	Cowal Gold Mine (CGM)	520
6.1	Introduction	520
6.2	Initial instability of the East high wall	520
6.3	Geology and geotechnical domains	520
6.4	Geotechnical considerations for the soft oxide zone	520
6.5	Hydrogeological results	522
6.6	Prediction of pore pressures	522
6.7	Depressurisation response	523
6.8	Dewatering well pumping and recovery test	524
6.9	Horizontal drainage response example for ultimate slope depressurisation	525
7	Whaleback South wall	525
7.1	Dewatering of the main ore body	525
7.2	Conditions in the South wall	526
7.3	Interpretation	528
8	La Quinua, Peru	528
8.1	General site setting	528
8.2	General mine dewatering	528
8.3	Phase 2C area	530
8.4	East high wall	530
8.5	Remediation plan	531
	Appendix 5 Cases studies for numerical modelling	536
	Case study 1: Numerical modelling of the North-west wall of the Diavik A154 pit	536
	1.1 Analysis of the DFN	536
	1.2 Use of the EPM Model	548
	1.3 Conclusions	555
	Case study 2: Numerical modelling of the marl sequence at the Cobre Las Cruces Mine, Andalucía, Spain	556
	2.1 Background	556
	2.2 Model development	558
	2.3 Conclusions	567
	Appendix 6 The lattice formulation and the <i>Slope Model</i> code	568
	1 The lattice formulation	568
	2 Features of the lattice approach	569
	3 Example application	570
	4 Validation of <i>Slope Model</i> using experimental data from Coaraze test site	573

4.1	Geometry	573
4.2	Boundary conditions	573
4.3	Building the model	574
4.4	Simulation results	575
4.5	Conclusions	575

Appendix 7 Lessons learnt and basic guidelines to monitoring for general dewatering **578**

1	Introduction	578
2	Water level monitoring	578
2.1	Reasons for monitoring	578
2.2	Water level measurement – practical guidelines	578
2.3	Monitoring frequency	580
2.4	Location of piezometers	580
2.5	Monitoring of pumping rates	581
2.6	Monitoring for hydrochemical fingerprinting	582
3	Summary and conclusions	582
	Symbols	583
	Abbreviations	584
	Glossary	585
	References	588
	Index	594

Preface and acknowledgements

Guidelines for Evaluating Water in Pit Slope Stability (the ‘Water Book’) is an outcome of the Large Open Pit (LOP) project, an international research and technology transfer project on the stability of rock slopes in open pit mines. It is a follow-up to Chapter 6 in the LOP project’s previous publication *Guidelines for Open Pit Slope Design* (Read & Stacey, 2009), which outlined how the hydrogeological model is developed and applied in the design of rock slopes. It describes the outcomes of hydrogeological research performed by the LOP project since 2009 and has the objective of providing slope design practitioners with a road map that will help them decide how to investigate and manage water pressures in pit slopes.

The LOP project was initiated in April 2005 and is managed on behalf of CSIRO Australia by John Read, CSIRO Earth Science and Resource Engineering, Brisbane, Australia. Since 2005 the project has been funded by the following mining companies: Anglo American plc; AngloGold Ashanti Limited; Barrick Gold Corporation; BHP Chile; BHPBilliton Innovation Pty Limited; Corporacion Nacional de Cobre Del Chile (‘Codelco’); Compañía Minera Doña Inés de Collahuasi SCM (‘Collahuasi’); De Beers Group Services (Pty); Debswana Diamond Company; Newcrest Mining Limited; Newmont Australia Limited; Ok Tedi Mining Limited; Technological Resources Pty Ltd (the RioTinto Group); Teck Resources Limited; Vale; and Xstrata Copper Queensland. Debswana and Collahuasi ceased to be sponsors in 2010 and 2011 respectively, and Xstrata Copper Queensland ceased to be a sponsor in 2012; AngloGold Ashanti and Ok Tedi Mining joined the project in 2010, and Teck joined in 2011.

In addition to the research performed by the LOP project since 2009, principally by Itasca International Inc., Schlumberger Water Services, and CSIRO Earth Science and Resource Engineering, the book draws on the experience of the sponsors and a number of industry consultants and practitioners who have shared their knowledge and experience by contributing to several of the sections in the book. In particular, the efforts of the following people are gratefully acknowledged.

- Enrique Buschiazzi, Schlumberger Water Services, Lima, Peru
- Ned Clayton, Schlumberger Water Services, Tucson, Arizona, USA
- Sergio Cosio, Kumtor Operating Company, Kyrgyzstan
- Branko Damjanac, Itasca Consulting Group, Inc., Minneapolis, USA
- Christine Detournay, Itasca Consulting Group, Inc., Minneapolis, USA
- Frederic Donzé, Formerly CSIRO Earth Science and Resource Engineering, Brisbane, Australia
- Greg Doubek, Doubek Hydrologic Inc., Denver, USA
- Blair Douglas, BHPBilliton Iron Ore, Perth, Australia
- Jeremy Dowling, Schlumberger Water Services, Denver, USA
- John De Souza, AngloGold Ashanti, Johannesburg, RSA
- Greg Ghidotti, RioTinto Group, Superior, Arizona, USA
- Bart Harthong, Formerly CSIRO Earth Science and Resource Engineering, Brisbane, Australia
- Mark Hawley, Piteau Associates, Vancouver, Canada
- Jim Hazard, Itasca Consulting Group, Inc., Minneapolis, USA
- David Holmes, Schlumberger Water Services, Shrewsbury, UK
- Peter Knight, Newcrest Mining Limited, Lihir, Papua New Guinea
- Richard LeBreton, Diavik Diamond Mines Inc, Yellowknife, Canada
- Chris Lomberg, Aquastrat, Perth, Australia
- Paul Mansfield, VAI Groundwater Solutions, Santiago, Chile
- Gordon MacLear, AngloGold Ashanti, Dakar, Senegal
- Rowan McKittrick, Schlumberger Water Services, Shrewsbury, UK
- Peter Milmo, RioTinto Iron Ore, Simandou, Guinea
- Rory O’Rourke, Datum Monitoring, Bury, UK
- Brian Peck, Schlumberger Water Services, Reno, Nevada, USA
- Manuel Rapiman, Minera Escondida Ltd., Antofagasta, Chile
- Mark Raynor, SRK, Shrewsbury, UK
- Ian Ream, Schlumberger Water Services, Tucson, USA
- Martyn Robotham, the RioTinto Group, Brisbane, Australia
- Jorge Rodriguez, Schlumberger Water Services, Perth, Australia
- Steve Rogers, Golder Associates, Vancouver, Canada
- Michael Royle, SRK, Vancouver, Canada
- Patrick Rummel, De Beers, Canada
- Bob Sharon, Newmont Mining Corporation, Tucson, Arizona, USA
- Rod Smith, Smith Water Services, Vancouver, Canada
- Peter Stacey, Stacey Mining Geotechnical Limited, Vancouver, Canada
- Craig Stevens, RioTinto Technology and Innovation, Salt Lake City, Utah, USA
- Bob St. Louis, Newmont Mining, Elko, Nevada, USA

- Rick Tunney, Iam Gold, Suriname
- John Waterhouse, Golder Associates, Perth, Australia

The book has been edited by Geoff Beale (Schlumberger Water Services, Shrewsbury, UK) and John Read (CSIRO Earth Science and Resource Engineering, Brisbane, Australia), with the assistance of an editorial subcommittee comprising Ashley Creighton (the RioTinto Group,

Brisbane, Australia), Loren Lorig (Itasca International, Inc., Minneapolis, USA), Marc Ruest (De Beers Global Mining, Johannesburg, RSA) and Michael Price (Independent Consultant, Church Stretton, UK).

Geoff Beale and John Read
August 2013

INTRODUCTION

John Read, Geoff Beale, Marc Ruest and Martyn Robotham

1 Scope of LOP project hydrogeological studies

The presence of water has a detrimental effect on slope stability. Water pressure acting in the pore spaces, fractures or other discontinuities in the materials that make up the pit slope will reduce the strength of those materials, and may therefore have a large influence on the performance, safety and economics of a mining operation.

The past 10 to 15 years has seen a significant improvement in the understanding of mining hydrogeology. Most mines now incorporate groundwater and surface water programs at all stages of the mining cycle. The programs are driven by both engineering and mine economics, and by increasing regulatory constraints, environmental laws and safety awareness. However, there remains a tendency to apply over-simplified approaches for assessing the distribution of pore water pressures in closely jointed rock masses and other pit slope materials. There is increasing awareness that an inadequate understanding of pore water pressures leads to mine production losses, either from conservatively over-designed slopes (i.e. slopes that are flatter than necessary), or under-designed slopes that fail.

In response to these issues, in 2009, the LOP project commenced a number of detailed hydrogeological studies to assess the effect of pore water pressures on the stability of jointed rock. The studies were designed to develop an understanding of the flow process in rock masses at different scales, particularly those with poor permeability and/or limited hydraulic connectivity between fractures, joints and other discontinuities. The objective was to develop a methodology that would help geotechnical design engineers decide how to treat water pressures in their pit walls.

The primary considerations for the studies have been as follows.

- The scale at which the water pressures are important for reducing rock strength, and how this relates to the grid and mesh sizes used in most geotechnical analyses.
- Whether water pressures are important throughout the rock mass or only in the major discontinuities.
- Whether analysis of the water pressure distribution can consider the rock mass as an equivalent continuum, so the flow analyses can employ some form of equivalent porous-medium approach.
- How slope deformation and dilation of the materials may change the water pressures within the rock mass and what are the resulting changes in the permeability of the fracture network.

For the geotechnical design engineer, a fundamental decision is whether to compute the water pressures only in the individual structures that cut through the rock mass; whether to compute the water pressures only in the intact rock within the intervening rock bridges (including the mélange of lesser joints, microfabric and intact rock substance included within the rock bridges); or whether to compute them in both.

To address these considerations, the hydrogeological studies focused on three core questions.

1. ***How does a fractured rock slope depressurise***, which sought to define the role and the response of the various orders of fractures within the rock mass by:
 - a. reviewing the literature from other industries with interests in the same or similar questions (e.g. nuclear, hydrocarbon and construction);
 - b. analysing the structural and hydrogeological data from a number of large open pit mines to develop a conceptual model of how depressurisation occurred in fractured rock pit slopes for different rock types, alteration zones and structural settings, and in a range of different mine site environments.
2. ***How should depressurisation be simulated numerically***, which sought to formulate a standardised

modelling approach for water pressure simulation for a range of environments and settings, with guidelines for determining:

- a. the approach that should be used to construct, calibrate and run different types of numerical models;
 - b. the selection of appropriate model codes and procedures for each type of hydrogeological setting;
 - c. when it is appropriate to use an equivalent porous-medium flow code;
 - d. when it is necessary to use a fracture flow code;
 - e. whether to use a transient or steady-state modelling approach;
 - f. when it is necessary to use a coupled hydromechanical modelling approach;
 - g. how active drainage measures (e.g. wells, drains, tunnels) should be simulated within the model.
3. **How should pore water pressures be incorporated into the slope design model**, which sought to outline recommended procedures and formats for providing basic water pressure distributions and responses into limit equilibrium and numerical slope stability programs, and assessing the impact of water pressures on rock mass strength.

Initially, it was intended that the information obtained from these three questions would form the basis of a handbook detailing the hydrogeological procedures that should be followed when performing open pit slope stability design studies. However, it was realised that such a handbook would not be complete without supporting sections that outlined how water pressures are characterised and reported, how slope depressurisation systems are implemented and maintained, and how predicted versus actual conditions and behaviour are monitored and reconciled.

Following the initial research using the data from Diavik and the other case study sites, it also quickly became apparent that the wider site-scale hydrogeology (and, in particular, groundwater recharge) is the most important factor for the assessment of water in the pit slope design. The scope of these guidelines was therefore further widened to include the characterisation and assessment of the wider site-scale hydrogeology. Furthermore, for a large number of pits, it is not possible to achieve the required pit slope pore pressure goals without the operation of an efficient general mine dewatering system. The general principles for modelling and design of mine dewatering systems are therefore also included.

The book was specifically designed to follow up on Chapter 6, (the Hydrogeological Model) in the LOP project's book *Guidelines for Open Pit Slope Design* (Read & Stacey, 2009). The contents are divided into the six sections shown in Figure 1 (The hydrogeological assessment process), starting with a section that outlines

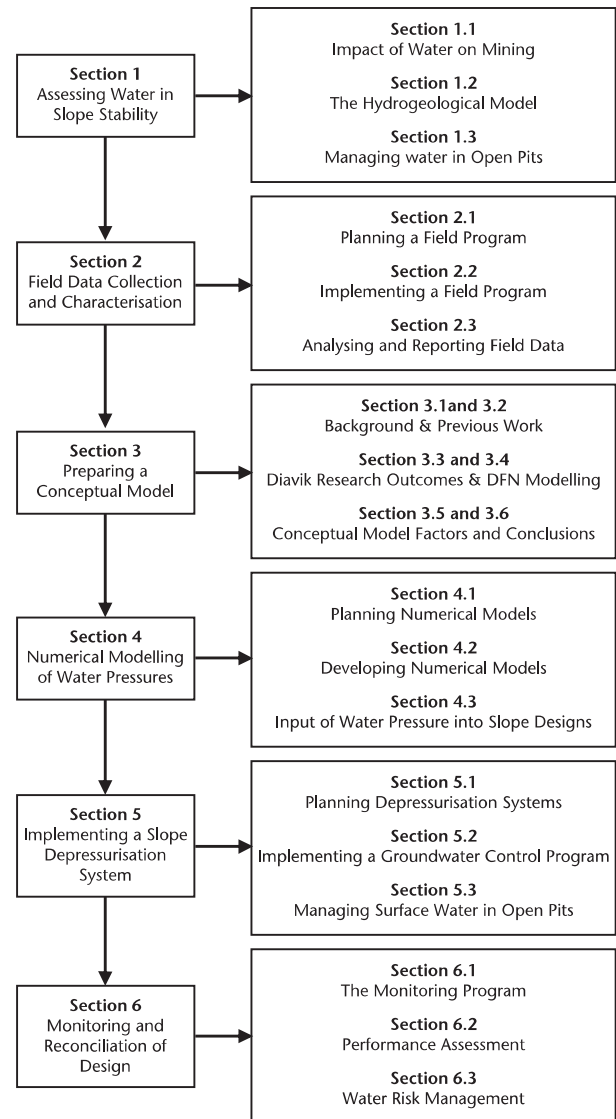


Figure 1: The hydrogeological assessment process

the framework used to assess the effect of water in slope stability (Section 1), followed by sections outlining how the water pressures are measured and tested in the field (Section 2), how a conceptual hydrogeological model is prepared (Section 3), how water pressures are modelled numerically (Section 4), how slope depressurisation systems are implemented (Section 5), and how the performance of a slope depressurisation program is monitored and reconciled with the design (Section 6).

The hydrogeological pathway followed through the slope design process (Figure 2) is illustrated in Figure 3. Sections 1, 2 and 3 are part of the Hydrogeology and Geotechnical Model sections of the process, Section 4 reports to the Groundwater Pressures and Stability Analyses sections, and Sections 5 and 6 relate to the Implementation section (general mine dewatering, slope depressurisation and monitoring).